Verify, Then Vote
Supporting Material

Grainulation

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This document accompanies *Verify, Then Vote: Overturning Wrong Majorities in Multi-Agent LLMs*. The main paper contains the rule, its harm argument, the empirical findings, and their limitations. This supporting material preserves the implementation details, formal proof catalog, evaluation methods, exploratory diagnostics, and reproduction instructions.

For the evaluation protocol and validation-channel audit, see Section S8. For reproduction commands, see Section S7. The scoring reducer and proof catalog are in Sections S2 and S4; sealed evidence and replay details are in Section S9. The full replay matrix and tamper-case inventory are in Section S10. Supporting sections, tables, and figures use an S prefix. Named links to the main paper refer to the separate `paper.pdf`; keep it alongside `supplement.pdf` for local navigation. Both documents are available from the paper repository.

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S1 Background

Staged compilation. LLVM separates compilation into front end, optimization passes over a shared IR, and back end (Lattner and Adve, 2004). The design adopts the same shape: agent outputs are source languages, the claim IR is the shared representation, and passes perform validation, conflict detection, and selection. As in compilers, a well-chosen IR makes the downstream passes simple. Here, typed triples make conflict detection a slot lookup.

Consensus and replicated data. Paxos (Lamport, 1998) and Raft (Ongaro and Ousterhout, 2014) achieve agreement among crash-prone processes, and PBFT (Castro and Liskov, 1999) tolerates f Byzantine faults with $3f + 1$ replicas. Our problem is different: agent failures are epistemic (wrong, stale, or miscalibrated claims) rather than crashes or protocol deviations, and the reducer decides once over a closed set rather than agreeing over time. From conflict-free replicated data types (Shapiro et al., 2011) the reducer borrows idempotent merge: re-submitting an identical contribution yields the same content-addressed identifier, and validation removes the duplicate.

Ensembles and frameworks. Dietterich (2000) gives statistical, computational, and representational reasons why ensembles outperform their members. Our design draws on the intuition but combines discrete claims under an explicit, recorded rule rather than averaging predictions. Orchestration frameworks for LLM agents route messages and tools between agents but leave the final decision to a model. Learned rankers likewise combine many signals through functions that are opaque at decision time. The reducer's distinguishing property is that each decision carries a per-feature justification that can be recomputed.

Voting theory. Arrow (1951) shows that no ranked voting rule over three or more alternatives satisfies unrestricted domain, non-dictatorship, Pareto efficiency, and independence of irrelevant alternatives together. Condorcet (1785) introduced pairwise majority comparison and the jury theorem: if voters are independently more likely than not to be right, majority accuracy approaches one as their number grows. The independence assumption is exactly what correlated LLM errors violate. Schulze's method (Schulze, 2011) is monotonic, clone-independent, and Condorcet-consistent. The reducer uses it as a tie-breaker over features rather than over agents, which Section S6.1 shows lets an unweighted feature vote.

S2 The Scoring Reducer

This section specifies the general reducer that verify-then-vote's selector sel runs on (main-paper rule): how answers become typed claims, how conflicts are detected, how candidates are scored, and when its tie-break provably cannot change correctness. On OVERTURN the simplest selector, plurality among verified answers, matches the scored one on accuracy (main-paper passage controls); the scoring reducer is the general form for open-ended claims with many slots per answer.

S2.1 Pipeline and running example

Figure S1 shows the six passes. Intent normalization (P1) produces the goal and criteria; if it uses a model, its output is an artifact like any other. Lowering (P2) is rule-based: clause segmentation, a 125-entry hedge lexicon for confidence, a rule cascade that extracts triples with polarity, evidence-marker patterns, and a round-trip reconstruction that grades each contribution HIGH, MEDIUM, or LOW. Validation (P3) drops contributions without a triple, rejects synthesis

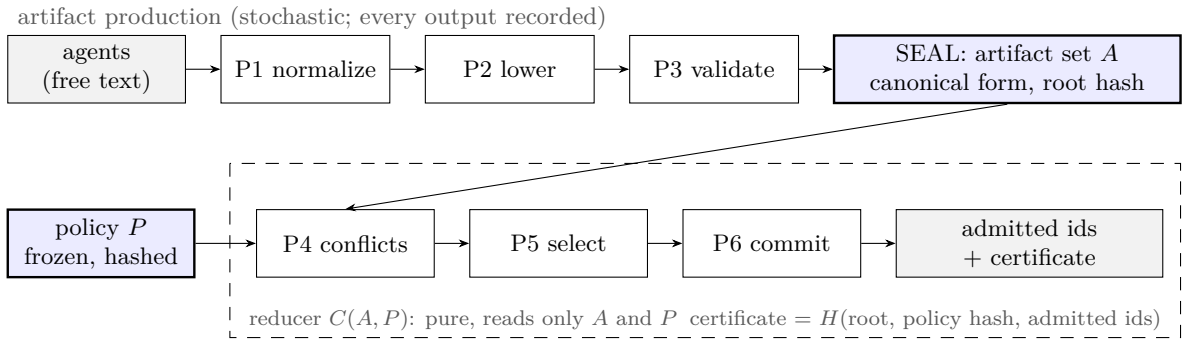


Figure S1: The arbitration boundary. Passes P1–P3 produce artifacts and may involve models; their outputs are sealed into A. Passes P4–P6 form the reducer C, which reads only A and P.

contributions (Section S2.3), orders the rest canonically, and removes duplicate identifiers. The reducer then detects conflicts (P4), selects (P5), and commits (P6).

Before the full example, we define the *deadband*: scores close enough to the highest score are treated as tied for a separate selection step. With a width of 0.05 and a top score of 0.60, a score of 0.58 enters that step, whereas 0.50 does not. The implementation uses quantized scores and a strict inequality, specified in Sections S2.4 and S2.6.

A small example shows the path. Asked for the capital of Australia, five agents answer in arrival order A–E: A says Sydney without evidence; B and C say Canberra, citing Wikipedia and the CIA World Factbook; D and E say Canberra without citations. All state confidence 0.8 under the default calibration factor 0.5. Because `has_capital` is functional, the five triples compete for one slot. The default policy scores them A 0.480, B 0.553, C 0.566, D 0.518, and E 0.530: citations add 0.06 and arrival order up to 0.05. C scores highest, but B, D, and E lie within the deadband of width 0.05 below it, so the tie-breaker runs over the four Canberra claims and selects B. Sydney and the other Canberra copies are rejected and kept in the audit record. Since every deadband candidate is correct, the tie-break decides only which copy is admitted. Yet the weakest correct claim (0.518) leads the incorrect one (0.480) by less than 0.05, so the separation condition of Section S2.4 fails even though the selection is safe: separation is sufficient rather than necessary.

S2.2 Claims and slots

A contribution carries an agent identifier, a kind (ASSERTION or CRITIQUE), a canonical triple (subject, predicate, object, polarity AFFIRM or NEGATE), a confidence, evidence items, its raw text, and its lowering grade. Its identifier is the first 128 bits of a SHA-256 hash over a canonical serialization of exactly these fields plus a scheme tag. The arrival index is excluded, since ingestion rather than the agent assigns it, but the root hash covers it. Sealing accepts only identifiers that equal their content hash, and replay re-checks every one, so editing a claim changes its identity. Binding the agent identifier into the hash gives two agents with identical text distinct identifiers; the agent identifier is an unauthenticated label.

Slots define what a conflict is, and their key depends on predicate arity. A *functional* predicate such as `has_capital` admits one object per subject, so its key is (subject, predicate) and different objects compete. A *non-functional* predicate such as `uses` admits many, so its key includes the normalized object and compatible claims occupy separate slots, within which AFFIRM and NEGATE still conflict. Keying by (subject, predicate) alone would make "TCP uses sequence numbers" and "TCP uses acknowledgments" compete for one occupant. Arity alone leaves exclusive objects of a non-functional predicate, such as a two-way versus a three-way handshake, indistinguishable from compatible ones, so the policy declares an *exclusivity table* of object groups; a grouped object is keyed by its group, whose members compete. Conflict detection (P4)

is thus a lookup over the declared arity and exclusivity tables (main-paper limitations).

S2.3 Output and provenance

The reducer's output is a set of contribution identifiers: the admitted/rejected partition over submitted ASSERTION contributions, plus each slot's members, scores, and winner. Commit (P6) admits each slot's winner and rejects the other members, which stay in the audit record; uncontested contributions are admitted unless they are critiques or LOW-graded, and a claim fewer than two agents asserted is flagged as uncorroborated. Because validation rejects SYNTHESIS contributions, and a property test checks that every admitted identifier is a submitted ASSERTION, provenance holds by construction: every output identifier names content some agent submitted. The reducer outputs identifiers rather than rendered text, and selection is embedding-free.

S2.4 Selection and its guarantees

The question for this section is when a tie-breaker cannot change the correctness of the reducer's choice. The answer depends on the combined score rather than any one feature. For example, two HIGH-graded candidates identical except for evidence values of 0.90 and 0.80 differ in total score by $0.35(0.90 - 0.80) = 0.035$ under the default evidence weight. That difference is smaller than the deadband width of 0.05, so both candidates proceed to the tie-break step. We now define all five score components and the conditions under which the deadband contains only correct candidates.

Scoring. Each candidate c has a feature vector $x(c) = (\mathrm{ev}, \mathrm{conf}, \mathrm{cs}, \mathrm{rec}, \mathrm{ca}) \in [0,1]^5$ and a lowering factor $\lambda(c)$, with $\lambda = 1.0, 0.85, 0.5$ for HIGH, MEDIUM, and LOW. Its score is

$$\mathrm{Score}_w(c) = \lambda(c)\, w^\top x(c), \qquad w = (0.35,\ 0.20,\ 0.25,\ 0.05,\ 0.15) \text{ by default.}$$

Evidence is $\mathrm{ev}(c) = \min\big(1, \sum_e w_e\, t_e\, q_e\, r_e\, m_e\big)$: stated weight, a type weight (tool result 1.0 down to reasoning 0.3), a verification factor (0.40 for unverified citations), a reachability factor (0.75 if marked reachable, 0.45 if not), and a cross-reference multiplier from 0.9 to 1.2 that grows with the number of agents citing the source. By default this checks only that evidence exists; Section S9 adds whether it supports the claim. *Confidence* is stated confidence times the agent's calibration factor. *Critique survival* is $\max(0, 1 - \sum_k \mathrm{severity}_k/5)$ over NEGATE critiques of the claim's triple, capped per agent. *Recency* is the rank of σ in the conflict set, scaled to $[0,1]$. *Criterion alignment* is the fraction of acceptance criteria the claim matches lexically.

Deadband and tie-break. When the top two quantized scores differ by less than $\delta = 0.05$, the reducer runs a Schulze widest-path election (Schulze, 2011) over the *deadband set* of candidates strictly within δ of the top, with each of the five features as a voter that prefers one candidate to another when its feature value is higher by more than an indifference threshold (0.01 by default); residual ties go to the earliest candidate in canonical order (lowest σ, then identifier). The policy offers order-free alternatives: zero recency weight, recency removed from the voters, residual ties ordered by identifier, a "highest score, then canonical order" tie-break, and value-level candidates. With the first three set, selection is independent of arrival order.

Why the margin must cover the total score. One might require only that correct claims carry more evidence than incorrect ones by more than δ. That condition leaves the tie-breaker exposed. Take two HIGH-graded candidates under the default weights with confidence 0.5, critique survival 1, and criterion alignment 1; the correct one has evidence 0.90 and arrived first,

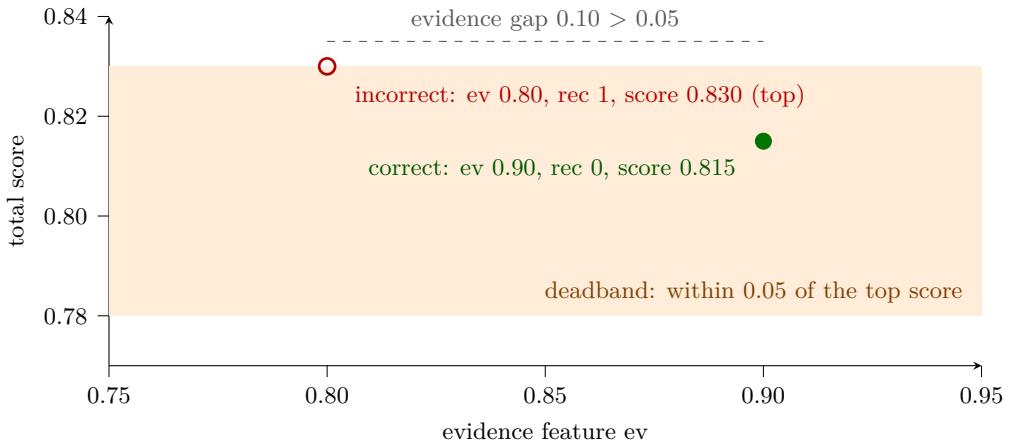


Figure S2: Why an evidence gap is insufficient. The correct claim has more evidence, but recency lifts the incorrect claim above it; both lie inside the deadband, so the tie-breaker may pick either.

the incorrect one evidence 0.80 and arrived last. Their scores are

$$\begin{aligned} \text{correct:} \quad & 0.35(0.90) + 0.20(0.5) + 0.25 + 0 + 0.15 = 0.815, \\ \text{incorrect:} \quad & 0.35(0.80) + 0.20(0.5) + 0.25 + 0.05 + 0.15 = 0.830. \end{aligned}$$

The evidence gap is $0.10 > \delta$, yet both candidates fall in the deadband and the incorrect one scores higher (Figure S2). The other four features and the lowering factor can outweigh any evidence gap, so a correct condition must be stated over the total score.

Notation. For a conflict set S_k whose correct and incorrect classes S_k^+ and S_k^- are both non-empty, and a pair $(c^+, c^-) \in S_k^+ \times S_k^-$, let $d(c^+, c^-) = \lambda(c^+)\, x(c^+) - \lambda(c^-)\, x(c^-) \in [-1, 1]^5$, so that $\text{Score}_w(c^+) - \text{Score}_w(c^-) = w^\top d(c^+, c^-)$.

Definition 1 (Score separation)**.** S_k is *score-separated* under w with margin $\delta \geq 0$ if $\Delta_w(S_k) = \min_{S_k^+} \text{Score}_w - \max_{S_k^-} \text{Score}_w > \delta$, equivalently if $w^\top d(c^+, c^-) > \delta$ for every pair.

Proposition 1 (Deadband homogeneity)**.** *If S_k is score-separated under w with margin $\delta \geq 0$, then (i) every maximizer of Score_w over S_k is correct, and (ii) every member of the deadband set $D_\delta = \{c : \text{Score}_w(c) \geq \max \text{Score}_w - \delta\}$ is correct.*

Proof. (i) If a maximizer $c^\star$ were incorrect, any correct c^+ would satisfy $\text{Score}(c^+) > \text{Score}(c^\star) + \delta \geq \text{Score}(c^\star)$. (ii) Every incorrect c^- has $\text{Score}(c^-) < \min_{S_k^+} \text{Score} - \delta \leq \max \text{Score} - \delta$.

Proposition 1 is close to definitional; its value is in naming what must be measured: the margin of the total score after lowering factors, not the evidence gap. When does evidence deliver that margin?

Proposition 2 (Evidence with bounded side-channels)**.** *Let $w \geq 0$, let $\gamma \leq \min d_1(c^+, c^-)$ bound the smallest lowering-weighted evidence gap from below, and for $i = 2, \ldots, 5$ let $\beta_i \geq \max(0, -d_i(c^+, c^-))$ bound from above the largest advantage an incorrect candidate holds on feature i. If $w_1\gamma - \sum_{i=2}^5 w_i\beta_i > \delta$, then S_k is score-separated with margin δ.*

Proof. For every pair, $w^\top d \geq w_1\gamma - \sum_{i \geq 2} w_i\beta_i > \delta$.

Without knowledge of the side-channels ($\beta_i = 1$), the condition becomes $w_1\gamma - (1 - w_1) > \delta$, which under the default $w_1 = 0.35$ and $\delta = 0.05$ requires $\gamma > 2.0$. Since $\gamma \leq 1$, **no evidence gap alone guarantees separation under the default weights**. Separation depends on the side-channels being small in practice, which is a property of the data that we measure (Section S5

Table S1: Per-citation factor $q \cdot r \cdot s$ for web evidence under the evidence policy.

Citation	$q \cdot r \cdot s$
Live page that supports the claim	0.638
Live page with no extractable text	0.525
Live page, not-enough-information	0.319
Unchecked	0.300
Dead or fabricated URL	0.090
Refuted	0

and Section S6.1). In the counterexample, $\gamma = 0.10$ and $\beta_{\text{rec}} = 1$ give $0.035 - 0.05 = -0.015$, exactly the observed score difference. Robustness to the weights has an equally exact answer.

Corollary 1 (Weight robustness over a polytope)**.** *Let $W = \operatorname{conv} V$ for a set of weight vectors V. S_k is score-separated with margin δ for every $w \in W$ if and only if it is for every $w \in V$.*

Proof. $\Delta_w(S_k)$ is a minimum of linear functions of w, hence concave, and a concave function on a polytope attains its minimum at a vertex.

The vertices of the full simplex are the unit vectors, so separation for *all* weights requires a correct claim to beat every incorrect one by more than δ on every feature at once, recency included. Full-simplex robustness is therefore essentially unavailable, and a robustness claim must name its polytope: separation verified at finitely many weight profiles extends exactly to their convex hull, and only for statistics linear in w, which excludes false-positive counts.

Lemma 1 (Tie-breaker neutrality)**.** *Let a tie-breaker T receive a non-empty candidate set $X \subseteq D_\delta$ and return a member of X. If S_k is score-separated with margin $\delta \geq 0$, then $T(X)$ is correct, as is the score maximizer, for every indifference threshold.*

Proof. By Proposition 1(ii), $X \subseteq D_\delta \subseteq S_k^+$, and by Proposition 1(i) the maximizer is in S_k^+.

The reducer's strict, quantized deadband set lies inside D_δ and always contains the top candidate, so Lemma 1 covers it (Section S2.6). All of these statements are machine-checked in Lean 4, together with the default-weights impossibility, the counterexample, and quantized versions of Proposition 1 (Section S4).

Separation is stronger than correct selection: swap which candidate is correct in the counterexample and the reducer selects correctly with $\Delta_w = 0.015 < \delta$. Separation is a property of each domain's data and must therefore be measured per conflict set, as main-paper evaluation does.

Proposition 2 shows that separation needs evidence to carry signal. By default the reducer only knows that a citation *exists*. Section S9 seals what the citation *says*.

S2.5 Evidence weights

Both record types are sealed into A's evidence store, and an evidence policy uses them. Verification and reachability factors come from the capture outcome, and a support factor multiplies each item: 1.0 for supports, a configured factor for not-enough-information, 0 for refutes. A label applies only if its checker identity is on the policy's list of accepted checkers; other labels score as unchecked. Because that list is part of the hashed policy, the reducer enforces acceptance-before-use; validation before acceptance is the study or application's responsibility. The cross-reference multiplier counts distinct agents whose capture of the same canonical source is labeled supports, rather than agents who typed the same string. Table S1 gives the per-citation factors for web evidence.

Table S2: Components and their backing.

Component	Backed by
Claim IR, content-addressed ids, arity-aware slots, exclusivity table	unit tests (compatible claims in separate slots; exclusive values in one conflict set)
Validation, SYNTHESIS rejection, output provenance	property test: admitted ids are submitted ASSERTION ids
Scoring, deadband, Schulze tie-break	equivalence test against the reference algorithm; property test that the winner lies in the deadband
Selection guarantees (Section S2.4)	Lean 4 proofs (Section S4)
Sealing and replay	in-memory, from-file, and four-build cross-replay; tamper tests (main-paper replay tests)
Evidence capture, support checking, evidence policy	tool tests; checker validation (Table S6); replay and tamper tests on sealed records
Verified-overturn gate (main-paper rule)	Lean 4 proof of the harm bound (Section S4); tests that every gated overturn's winner is a witness; permutation invariance over 120 orderings; independent re-implementation matched on 7,910 OVERTURN slots
Rule-based lowering	worked examples (Section S3)

Separation on grounded evidence. Under the gate, the selector decides among the deciding stage's candidates only, the plurality set or the verified set (main-paper rule definition), so the tie-break neutrality lemma applies to the actual decision when that candidate set is score-separated over its own members or contains only correct members. On OVERTURN's 595 decisive conflicts of P and T this holds for 467 decisions (P 432 of 526, T 35 of 69), and the winner is correct on all 467: in 460 the gate leaves only correct candidates, and in 7 a mixed candidate set is separated by more than δ. Measured over all of a question's answers instead, without the gate's restriction, 229 of the 595 sets are separated.

S2.6 Implementation

The reducer is implemented in about 12,000 lines of Rust, with the evidence tools in Python. The reducer, sealing, and replay are covered by 132 unit and property tests; the evidence tools have 136 tests that run with networking blocked, including 26 direct server-side request forgery forms. Table S2 maps each component to what backs it.

All hashes are truncated SHA-256: root hash, policy hash, slot and contribution identifiers keep 128 bits, the certificate 64. Canonical serialization is sorted-key JSON in the spirit of RFC 8785 (Rundgren et al., 2020) with shortest round-trip floats, with its own number formatting rather than JCS's.

The Schulze tie-break is the only super-linear step. The reference algorithm computes Floyd–Warshall widest paths over each deadband set, cubic in its size. The implementation computes the same winner, ties included, by building widest-path trees lazily for each leader of the reference algorithm's sequential scan with a bucketed Dijkstra over at most six strength levels, switching to a bitset closure after a bounded number of leaders; the worst case is still cubic. A test compares winners with the reference algorithm on 24,000 random sets of 2 to 400 members across indifference thresholds and strategies.

The deadband uses a strict inequality on quantized scores, and the strictness is necessary: a real margin of 0.05004 can round to a quantized gap of exactly 500, so a non-strict deadband could admit an incorrect candidate under separation. A property test over 2,000 random sets checks that the Schulze winner lies in the deadband set. The Lean 4 formalization (with Mathlib, about 670 lines, 46 theorems, every goal proved) works in exact arithmetic, with floating-point error before quantization outside the model, and treats the tie-breaker abstractly, since Lemma 1 holds for any selector that returns a member of its input. The formalization shows both precon-

ditions, a non-negative deadband and a non-empty correct class, necessary by counterexample.

S3 IR Schema and Lowering Examples

The schema below follows the implementation; wire names are shown in code font. The lowering examples are hand-traced illustrations of the rule cascade, and their numeric values are illustrative.

```
Contribution {
  id:          first 128 bits of SHA-256 over the canonical form of
               (agent_id, kind, canonical_form, confidence, evidence,
               lowering_confidence, raw_text, scheme tag); excludes seq
  seq:         u64; recorded arrival index, inside the root hash
  agent_id:    String; a label bound into id; unauthenticated
  kind:        ASSERTION | CRITIQUE  (SYNTHESIS is rejected)
  canonical_form:      CanonicalTriple | null
  confidence:          [0, 1]
  evidence:            EvidenceItem[]
  raw_text:            String; original agent output
  lowering_confidence: HIGH | MEDIUM | LOW
}
CanonicalTriple { subject, predicate, object,
                  polarity: AFFIRM | NEGATE }
EvidenceItem {
  typ:          TOOL_RESULT | EMPIRICAL | CITATION
                | AUTHORITY | REASONING
  source:       String; URI, tool call id or reasoning text
  weight:       [0, 1]
  reachable:    bool; agent-reported
  verification: UNVERIFIED | REACHABILITY_CHECKED
                | CONTENT_HASHED | CROSS_REFERENCED | STALE
  capture_ref:  String?; id of a sealed capture record
  support_ref:  String?; id of a sealed support record
}
```

Slot key: `(subject, predicate)` for functional predicates; `(subject, predicate, normalized object)` for non-functional ones, or `(subject, predicate, group)` when the object belongs to a declared group in `incompatible_values`. Redundancy key: `(subject, predicate, object, polarity)`. The intent is a goal string with a list of acceptance `criteria`; the calibration snapshot is the sealed `calibration` map. Policy options in Section S2.4 have these wire names: recency-free tie-break voters, `schulze_dims = 23` (bitmask over ev, conf, cs, rec, ca; default 31); residual-tie order, `tiebreak_order = seq | id`; score-then-canonical, `tiebreak = score_then_canonical`; value-level candidates, `candidate_mode = value:max | value:pooled`; evidence policy, `evidence` with `accepted_checkers`.

Example 1 (medical). "Metformin reduces HbA1c levels by approximately 1–1.5% in type 2 diabetes patients, according to the ADA Standards of Care (2024)." One clause yields `(metformin, reduces, HbA1c levels by 1-1.5%)`, AFFIRM; the hedge "approximately" maps to a high confidence band (0.85); the citation becomes a CITATION item of weight 0.8; the lowering grade is HIGH.

Example 2 (legal). "Under GDPR Article 17, data subjects have the right to erasure, but this right is not absolute and can be overridden by legitimate interests." Segmentation

at "but" yields `(data subjects, have_right_to, erasure)` AFFIRM, `(right to erasure, is, absolute)` NEGATE, and `(right to erasure, can_be_overridden_by, legitimate interests)` AFFIRM, with one CITATION item.

Example 3 (technical). "TCP uses sequence numbers and acknowledgments to ensure reliable delivery. Lost packets are retransmitted after a timeout." The conjunction splits into `(TCP, uses, sequence numbers)` and `(TCP, uses, acknowledgments)`; because `uses` is non-functional, the two occupy separate slots and are compatible. The passive clause yields `(lost packets, are_retransmitted_after, timeout)`. The evidence is REASONING only.

Known failure modes. A double negative ("It is not uncommon for TCP connections to experience packet loss") is lowered with NEGATE polarity although its meaning is affirmative; the grade drops to MEDIUM but the triple is wrong. Modality is lost ("Vitamin D supplementation might reduce the risk of respiratory infections" becomes an unqualified `reduces`), with the hedge lowering confidence to about 0.55 but conflict detection treating it like an assertion. Temporal qualification is lost ("Pluto was classified as a planet until 2006…" yields `(Pluto, classified_as, planet)`), which creates a false conflict with agents asserting the dwarf-planet classification.

S4 Proof Catalog and Hash Definitions

All statements of Section S2.4 and of the gate (main-paper rule) are proved in Lean 4 (de Moura and Ullrich, 2021) with Mathlib (The mathlib Community, 2020); the build contains no `sorry`. In Lean, features are indexed from 0 (0 = ev, 1 = conf, 2 = cs, 3 = rec, 4 = ca), one less than in Section S2.4; results stated for an arbitrary score function over any linear ordered field apply to the reducer's idealized exact-arithmetic score model. The quantized-score results require the stated rounding-error bounds; floating-point arithmetic before quantization remains outside the formalization.

Statement	Lean theorem	Notes
Score difference equals $w^\top d$; $d \in [-1,1]^5$	`score_sub_eq_dot`, `dvec_mem_Icc`	the second assumes features and λ in $[0,1]$
Definition 1 and its min/max form	`Separated`, `separated_iff_min_sub_max`	needs both classes non-empty
Proposition 1 (i)	`prop1_i`	every maximizer; needs $\delta \geq 0$ and a non-empty correct class
Proposition 1 (ii), non-strict and strict deadband	`prop1_ii`, `prop1_ii_strict`, `inDeadbandStrict_imp`	
Necessity of $\delta \geq 0$ and of a non-empty correct class	`prop1_i_needs_nonneg_delta`, `prop1_needs_nonempty_pos`	counterexamples
Lemma 1, general and with the strict deadband set	`lemma1`, `lemma1_strict`	also needs X non-empty and $T(X) \in X$
Proposition 2	`prop2`	any lower bound γ and upper bounds β
Unit-β form; default weights require $\gamma > 2.0$	`prop2_condition_unit_beta`, `default_condition_iff_gamma_gt_two`	the second at $\delta = 0.05$
No $\gamma \leq 1$ suffices under default weights; even a maximal evidence gap fails to separate	`default_weights_no_gamma`, `default_max_evidence_gap_not_separated`	every $\delta \geq 0$
Corollary 1	`cor1`	both directions, any generating set V

Statement	Lean theorem	Notes
Full-simplex robustness iff every feature separates	`simplex_robust_iff`	
Proposition 1 on $\times$10,000 quantized scores	`prop1_i_quantized`, `prop1_ii_quantized`, `quantized_gap`	assumes rounding error at most $\frac{1}{2}$
A non-strict quantized deadband is unsafe	`quantized_nonstrict_fails`	margin 0.05004 rounds to a gap of 500
Counterexample: scores, evidence gap, incorrect wins, unseparated, an evidence gap above δ does not imply separation, Proposition 2 condition fails	`score_cPlus`, `score_cMinus`, `old_condition_holds`, `incorrect_wins`, `within_deadband`, `not_separated`, `old_prop1_false`, `prop2_quantities`	computed over $\mathbb{Q}$
Swapping correctness labels selects correctly without separation	`mirror_correct_but_not_separated`	swapping only recency values gives margin 0.085, which is separated (`recency_swap_is_separated`)
Main-paper harm proposition: a correct default lost by the gate implies a verified incorrect value	`Gate.gate_harm`	needs only selection closure, $\mathrm{sel}(S) \in S$ for non-empty S; correctness is a function of the value
The gated winner is verified whenever any value is	`Gate.gate_winner_verified`	
Main-paper harm corollary, counting form $c_D \leq F$	`Gate.gate_cD_le_F`	slot data and selector may differ per item
Main-paper harm corollary, no-loss, and its exact characterization	`Gate.gate_no_loss`, `Gate.gate_correct_iff`	when no incorrect value is verified
The plurality set is non-empty; the default and gated winners are slot values	`Gate.P_nonempty`, `Gate.D_mem_P`, `Gate.gated_mem_vals`	needs a non-empty slot

Outside the formalization: floating-point error before quantization (the model uses exact arithmetic), Schulze itself (Lemma 1 holds for any selector returning a member of its input), and the remark that false-positive counts are nonlinear in the weights.

Policy hash. SHA-256, truncated to 128 bits, over a canonical JSON object with two sections, sorted by key and written without whitespace: `config` (every configurable field, including the exclusivity table with each group's objects sorted and de-duplicated) and `constants` (score scale, lowering factors, evidence-type weights, verification table, and the sorted functional predicates). Floats use shortest round-trip decimal form; non-finite floats are rejected. Optional fields are serialized only when they differ from their defaults, so setting any of them changes the policy hash and therefore the certificate.

Root hash and certificate. The root hash is SHA-256, truncated to 128 bits, over the canonical serialization of A, including the identifier-scheme tag. The certificate is SHA-256, truncated to 64 bits, over the root hash, the policy hash, and the admitted identifiers in admission order.

S5 Synthetic Workload

How does the reducer behave as inputs grow? A generator draws contributions for three fixed topic banks (Australia's capital, factual; TCP reliability, technical; saturated fat, contested) from

Table S3: Synthetic workload, one end-to-end compilation per cell: slots / false positives / precision / median time over five runs (release build, single-threaded, seed 42, Apple M3 Pro, shared machine).

N	Australia	TCP	Saturated fat
10	4 / 0 / 100% / 0.4 ms	9 / 3 / 67% / 0.3 ms	7 / 1 / 86% / 0.3 ms
100	4 / 0 / 100% / 2.7 ms	14 / 3 / 79% / 2.7 ms	15 / 5 / 67% / 3.0 ms
1,000	4 / 0 / 100% / 24.2 ms	14 / 3 / 79% / 28.3 ms	15 / 5 / 67% / 26.9 ms
5,000	4 / 0 / 100% / 128 ms	14 / 3 / 79% / 135 ms	15 / 5 / 67% / 140 ms
10,000	4 / 0 / 100% / 273 ms	14 / 3 / 79% / 284 ms	15 / 5 / 67% / 305 ms

Table S4: Measured total-score margin Δ_w on synthetic contradiction sets.

Topic (contradiction sets)	$N = 100$	$N = 1{,}000$	$N = 10{,}000$
Australia (1)	separated (0.051)	below δ (0.004)	inverted (-0.012)
TCP (3)	2 separated; 1 at 0.036	all inverted (min -0.191)	all inverted (min -0.237)
Saturated fat (1)	separated (0.170)	below δ (0.042)	below δ (0.022)

20 agent personalities, emitting triples directly. It assigns correctness, evidence, and lowering grade by construction: correct claims get citation or empirical evidence of weight 0.80–0.95, misconceptions reasoning-only evidence of weight 0.2–0.4 or none. The workload therefore tests whether the scorer uses the signals it is given, which cooperative agents supply and adversaries would withhold.

Time grew about 10–11× from $N = 1{,}000$ to $N = 10{,}000$ as claims grew about 10×. The deadband is entered in nearly every conflict set from $N = 100$ upward and its sets grow with N, which is why the Schulze implementation matters: the reference algorithm took 7.50 s for Australia at $N = 10{,}000$, 27× longer (Runtime, below). These are workload measurements rather than complexity bounds, and the stable slot counts reflect each bank's fixed size. They do establish that arbitration is cheap at realistic sizes: a whole 100-agent compilation takes about 3 ms, while the five agents of Section S6.1 took a median of 48.3 s (range 15.6–79.1 s) to answer one question, more than four orders of magnitude longer. On OVERTURN the same holds per decision: 6 ms of reduction (main-paper audit and cost results) against a median of about 25 s for the five agents to answer an item.

Two selection findings carry over to real data. Most false positives are uncontested: at $N = 100$, all 3 TCP and all 5 saturated-fat false positives were admitted without any conflict, such as "guarantees zero data loss," which puts them beyond any reducer's reach. And separation erodes as N grows while the winners stay correct (Table S4), exactly the case Section S2.4 anticipates: at large N correct winners come from the score distribution, not the guarantee.

Runtime. On the three banks the claim count grew from 1,561 to 15,743 for Australia between $N = 1{,}000$ and $N = 10{,}000$. From $N = 100$ upward the deadband was entered in 4 of 4 Australia, 13 of 14 TCP, and 15 of 15 saturated-fat conflict sets, and at $N = 10{,}000$ the Australia deadband sets held 967–1,148 candidates. The reference Schulze algorithm, timed in a single run, took 7.50 s, 1.40 s, and 1.95 s on the three banks at $N = 10{,}000$, against 273 ms, 284 ms, and 305 ms for the implementation: 27× for Australia and 5–6× for the others, with identical outputs. In a per-pass breakdown for Australia at $N = 10{,}000$ (median total 316 ms), selection took about 58 ms, of which Schulze about 26 ms over four deadband sets of 967–1,148 candidates, and commit dominated at about 196 ms.

Baselines ($N = 100$, 50 seeds; mean false positives / mean precision).

Method	Australia	TCP	Saturated fat
Reducer	0.0 / 100%	2.7 / 80.3%	5.0 / 66.8%
Majority vote	0.0 / 99.0%	3.2 / 76.5%	5.0 / 66.8%
Best confidence	0.2 / 95.0%	2.8 / 79.7%	5.2 / 65.3%
Last writer	0.2 / 95.0%	3.3 / 76.0%	5.2 / 65.2%
Random	0.5 / 86.5%	3.7 / 72.8%	5.2 / 65.1%

Because the generator gives correct claims stronger evidence by construction, a small advantage for an evidence-weighted reducer is expected by design and confirms only that the scorer uses its inputs.

Adversarial-ratio sweep (seed 42; 0, 20, 40, 60, 80, and 90% adversarial agents, drawn from a separate eight-template roster, which is why TCP has 2 false positives at 0%). TCP false positives were 2 at every ratio for the reducer and for best-confidence, 2, 2, 2, 2, 4, 4 for majority vote, and 2, 3, 3, 2, 4, 4 for random selection. Australia had 0 for every method except majority and random at 90% (1 each); saturated fat had 4–5 for all methods. These adversaries are generator archetypes with weak or no evidence, so the sweep tests weak-evidence adversaries; source misrepresentation is the separate threat of main-paper setting.

Ablation (ten profiles: default, each single weight zeroed and the rest renormalized, evidence-only, confidence-only, critique-only, uniform). False-positive counts stayed the same under every profile (Australia 0, TCP 3, saturated fat 5). The three single-component profiles are three of the five simplex vertices, but the convex-hull extension of Corollary 1 applies only to statistics linear in the weights, which excludes false-positive counts. Which copy wins does change: with evidence zeroed, 3 of 4 Australia winners differ.

S6 Free-Text Measurements

Results. The five models answered the 55 questions twice: Collection 1 dispatched requests sequentially in a fixed model order, and Collection 2 concurrently in seeded random order with arrival recorded. Table S5 gives every method on both collections under both graders, and Figure S3 the total-score margin of every decisive set. A question-clustered bootstrap (10,000 resamples) of the reducer's accuracy minus majority's gives −0.15 [−0.32, 0.00] on Collection 2 and 0.33 [0.09, 0.57] on Collection 1 under the lexical grader, and 0.00 [−0.22, 0.27] and 0.06 [−0.18, 0.33] under the polarity-aware grader. On the concurrent collection, 9 of 14 decisive sets fell in the deadband and the latest arrival won 7 of those 9. Sealing left every selection unchanged across all 926 conflict sets of the sequential collection: the boundary is accuracy-neutral by design.

Graders. The lexical grader matches by token overlap with the concatenated gold-triple text, checks predicates only when object overlap is below 0.6, and ignores polarity, so "Pluto is not classified as a planet" matches `(Pluto, is_classified_as, dwarf planet)`. The polarity-aware grader requires subject, object, one of 17 predicate synonym groups, and matching polarity, scoped per clause; it has its own misses, for example "299,792 km/s."

Evaluation-harness cross-check. Collection 1 was also evaluated with a separate Python harness of the same scoring form but different details: a flat source factor instead of verification and reachability, criterion alignment and critique survival fixed at 1, calibration read from the responses, recency from model-list position, polarity ignored, and selection without deadband or Schulze. Its selections score 20 of 24 under the lexical grader and 12 of 17 under the polarity-aware grader, the figures of Section S6.1; disabling its context-aware citation association removed

Table S5: Correct selections over eligible · decisive conflict sets. "vs maj." gives the discordant pairs b/c against majority vote and the exact McNemar p. Column headers give the eligible / decisive counts.

	Collection 2			Collection 1		
Method	lexical (26 / 14)	vs maj.	polarity-aware (16 / 9)	lexical (24 / 15)	vs maj.	polarity-aware (17 / 9)
Reducer, default policy	18 · 6	2/6, 0.29	12 · 5	20 · 11	9/1, 0.021	12 · 4
Recency weight 0	18 · 6	2/6, 0.29	12 · 5	20 · 11	9/1, 0.021	12 · 4
Recency weight 0, no deadband	19 · 7	0/3, 0.25	11 · 4	15 · 6	4/1, 0.38	12 · 4
Evidence only	19 · 7	0/3, 0.25	11 · 4	15 · 6	4/1, 0.38	12 · 4
Majority vote	22 · 10	—	12 · 5	12 · 3	—	11 · 3
First member (earliest arrival)	22 · 10	—	12 · 5	13 · 4	—	11 · 3
Last member (latest arrival)	16 · 4	—	11 · 4	16 · 7	—	11 · 3

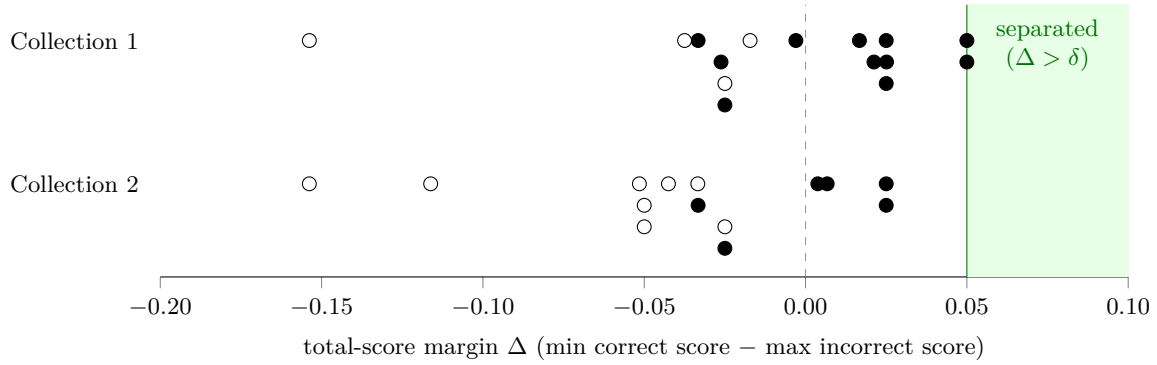


Figure S3: Total-score margin of every decisive conflict set in the free-text study (lexical grader). Filled: the reducer selected correctly; open: incorrectly. Every set lies below the separation region; many correct selections have negative margins. Synthetic margins are in Table S4.

a further 3 of the 20 correct selections, and that association changed the selection in 8 of 926 sets. The sealing control covered the reducer, evidence-only scoring, and majority vote alike. A Python re-implementation of the implemented reducer's detection and selection matched the Rust binary on all 55 questions; the harness and the implemented reducer agreed on 791 of 860 Collection 2 conflict sets and 861 of 926 Collection 1 sets, and every disagreement had an identified cause (rejection of LOW-only singletons, non-functional predicates, and one deadband tie-break per collection).

Failure analysis and precision. In the harness's four incorrect selections on Collection 1, all five claims carried reasoning-only evidence (vaccines); 11 single-agent claims tied on evidence and recency picked the last (goldfish); the correct claim had MEDIUM lowering (an evolution question); or the subject was malformed. Only one set had cited evidence on both sides. Macro precision was low for every method (lexical: reducer 0.216, majority 0.199; polarity-aware: 0.087 and 0.082). A heuristic sort of the 871 reducer winners failing the polarity-aware grader gave 439 extra-but-true, 354 extraction errors, 77 grader misses, and 1 incorrect, itself a grader miss; these categories are unadjudicated, so low precision may reflect incorrect outputs, incomplete gold, or grading errors in unknown proportion.

Canonicalization. The gold-blind object canonicalizer normalizes markdown, case, punctuation, articles, plurals, number formats, units, and approximation hedges, with a two-entry synonym table as its only semantic step. It dissolved 3 of 66 Collection 2 contradiction sets (2 of 63 in Collection 1), none decisive; on Collection 2 eligible sets fell from 26 to 23 and decisive sets stayed at 14, with majority equal to first-member on 23 of 23 and 14 of 14; decisive sets

whose top object occurs more than once rose only from 1 to 2. Across all eight table, grader, and collection cells, a single decisive outcome changed.

Value-level candidates. Treating each distinct value of a slot as one candidate, aggregated by best member or by evidence pooled across distinct agents, changed almost nothing: best-member left every winning value unchanged on both collections, pooling changed one or two, and Table S5 accuracies were unchanged except one extra correct pooled selection on Collection 1 under the polarity-aware grader. Only about 10% of slots contain a repeated value. On the synthetic workload, value mode matched or underperformed member mode: best-member lost 1 of 3 TCP conflicts at $N = 10{,}000$, and pooling lost all 3 at $N = 1{,}000$ and $N = 10{,}000$, because the capped evidence sum saturates for both values.

S6.1 Four requirements for evaluating arbitration

Score-based arbitration can look better or worse than it is for reasons unrelated to its decisions. We identify four such distortions and the protocol element that removes each. Each was measured on the same five models answering 55 factual, technical, contested, and misconception questions in free prose with sources and confidence (Section S6). OVERTURN meets all four, and they apply to any evaluation of arbitration, ours included.

Selection must be order-free. A per-feature tie-break lets arrival order vote even at zero recency weight, because each feature votes regardless of its weight. Of 308 deadband slots, removing recency from the voters changed the winner in 236, and with concurrent dispatch the latest arrival won 7 of 9 decisive deadband sets. The order-free policy sets recency weight to zero, removes it from the voters, and orders residual ties by identifier. Under it every OVERTURN decision is independent of arrival order. Breaking majority ties by earliest arrival instead of at random leaves the result at 90 corrections against 11.

Answers must share a closed space. Free-text objects rarely repeat verbatim, so majority vote reduces to the first-seen answer: its correctness equalled first-member's on all 26 eligible sets, with or without gold-blind canonicalization. An answer line over a closed space, canonicalized gold-blind, makes the majority count real agreement, so the baseline is the strong one.

Citations must be checkable. Open citations mostly cannot be verified. Of 275 responses, 129 cite 332 unique identifiers. Of these, 166 (50%) fail to resolve, 109 (33%) yield checkable text, and 3.8% occur in the sentence of the claim they would support. A fixed corpus with citations restricted to retrieved passages makes evidence checkable. On 84% of decisive conflicts from SciFact, PopQA, and FEVER, the retrieved passages include one the checker labels as supporting the gold answer. Score separation over all of a question's answers holds in 229 of the 595 decisive conflicts of P and T, against none of 29 on free text (Figure S3). The tie-break lemma covers 467 of the 595 actual gated decisions (Section S2.5).

Grading must respect polarity. A token-overlap grader that ignores polarity counts "Pluto is not classified as a planet" as matching the gold claim that Pluto is a dwarf planet. It manufactures an apparent advantage: 20 of 24 eligible sets for an evidence-weighted reducer against 12 for majority vote ($p = 0.021$). A polarity-aware grader reduces this to 12 against 11 of 17. OVERTURN grades against dataset gold with an answer key, exact match or PopQA's published alias rule, which two model annotators blind to its verdicts matched on 162 of 169 answers ($\kappa = 0.917$).

S7 Artifact and Reproduction

The study scripts and sealed records accompany the paper at `https://github.com/graination/papers/tree/main/papers/verify-then-vote`. The artifact README gives portable source-reconstruction commands and exact public-redaction mappings; these wrappers verify the recorded decisions without claiming original-binary identity. "Models" means the five locally served models with the recorded digests.

Result	What to run	Expected output
Lean proofs (Section S2.4, Section S4)	Lean build of the proof library	build succeeds with no `sorry`; the catalog maps each statement to its theorem
Unit and property tests, Schulze equivalence (Section S2.6)	reducer test suite (release build)	132 tests pass, including the 24,000-set equivalence and 2,000-set deadband properties
In-memory and from-file replay (main-paper replay tests)	quick test target; file-replay script	500 identical certificates; 5 of 5 topics replay; policy tampers rejected
Cross-build replay and tamper (Tables S8 and S9)	per-build dump and replay driver on each of the four builds, then the comparison script	byte-identical dumps per topic; 80 of 80 replays pass; 36 of 36 tamper cases rejected identically
Synthetic workload (Table S3, Table S4, Section S5)	reducer's stress, pass-timing, score-gap, baseline, and ablation modes, seed 42	Tables S3–S4 and Section S5 (timings vary by machine)
OVERTURN benchmark (Section S8 and main-paper accuracy results and main-paper audit and cost results)	benchmark build (from downloaded datasets) and retrieval corpus build; collection script (needs models); check, judge, and evaluation scripts; re-analysis from the sealed records needs neither models nor hosted calls	item files and corpus matching the freeze record; main-paper accuracy tables; harm-bound check; replay, leave-one-agent-out, and cost tables
Matched gate ablation (main-paper matched gate table)	`reproduction/matched_gate.py`; native reducer and sealed inputs, no model calls	four matched policies; 4,400 Rust/Python decision comparisons; main-paper matched gate table
Validation-channel audit (Section S8.2)	`reproduction/audit_validation.py`; frozen validation rows, no model calls	raw false supports 18/350, false refutes 22/350; 82 blank human-review forms, not completed judgments
Passage control (main-paper passage controls)	control script (calls the hosted checker); its call log and labels are released	0 of 236 pairs supported in each condition; 74.0% re-decided accuracy
Verified wrong answers (main-paper harm analysis)	per-pair classification table over the sealed captures and labels	82 pairs on 56 items, two annotators ($\kappa = 0.95$): 24 checker, 8 entity, 5 source, 19 gold (9 alias, 3 alternative, 7 source–gold); lost defaults 5, 1, 1, 3
Second-pool evaluation on the same questions (main-paper second-pool results, main-paper second-pool table)	qualification and collection scripts (need the second pool's models); check, judge, and evaluation scripts; verified-default-first decision script; re-analysis from the sealed records needs neither	freeze record matching all hashes; main-paper second-pool table; harm-bound check; pooled and design-set sensitivity analyses
Cost with label reuse (main-paper audit and cost results)	cost script over the sealed call logs of both pools (no model call)	\$0.021 per contested decision at original prices, \$0.011 in new calls; 49% of cited checks reused
Checker wording stability and two-way checker (Section S8)	paraphrase re-check script (calls the hosted checker) and two-way re-decision script	2.6–5.7% label flips against 4.4% re-query noise; 87.3% to 77.8% with a two-way checker
Support-checker validation (Table S6)	validation script (build, run, report; run calls the hosted model)	Table S6 metrics
Free-text study (Section S6.1, Section S6)	collection script (needs models), evaluation, table, capture, and attachment scripts	Table S5, Figure S3; 332 identifiers and their capture outcomes; attachment rates

Preregistration and deviations. The OVERTURN preregistration, its development-stage selections, the reducer binary, the evaluator, the checker identity, and the policy templates were frozen by hash (56 files) before confirmatory collection; all hashes were re-verified before the single confirmatory analysis. A pre-specified non-binding futility look at 300 items read only the corrected and overturned counts of the primary contrast (25 against 4). The local-checker arm (H2) is reported as not run, because the candidate local checker did not reach the validation thresholds before the freeze, and the local judge is exploratory. The client used for every hosted call was pinned to the version in the checker identity, and one check, one reader call, and one closed-book call returned refusals, scored as unchecked and as wrong respectively. Two judge analyses, regeneration and leave-one-agent-out, ran in fresh caches so that each call is new. The evaluator's per-decision check cost counts only calls made in its final step, because earlier calls were served from the shared label cache; the main-paper audit and cost section reports the cost from the complete call logs. The companion memory-only benchmark of main-paper passage controls (600 confirmatory items) was analyzed once with its frozen evaluator after OVERTURN was frozen, as its preregistration required; its amendment-5 analyses (non-inferiority and cost) use a script that imports the frozen test functions, its evidence-verification hypothesis was not run because no confirmatory capture exists, and its timing ran on a loaded machine. Item files and captured passages derived from FEVER keep its CC BY-SA 3.0 license, and other third-party data keep their own licenses, listed with the artifact; full datasets and corpora are represented by their hashes and fetch commands. Model digests for all five agents are recorded in the run manifests.

Second pool. Its preregistration, pool manifest, and freeze record (82 files) were frozen before confirmatory collection, with verified default first as the primary rule and the OVERTURN hypotheses repeated unchanged; the first pool's labels and its checker, judge, and policy identities are reused byte-identically. One candidate agent failed qualification because its model server stopped on every request, and its first pre-declared alternate replaced it. A two-way local checker considered as a secondary arm was removed before the freeze, because it cannot verify a REFUTED answer under the unchanged claim representation (Section S8). Four checks returned refusals and are scored as unchecked. The evaluator's per-decision check cost for the second pool ($0.0083) counts only labels first checked in its confirmatory run, not those from its futility look or reused from the first pool; the main-paper audit and cost section reports the cost from the complete call logs of both pools. The verified-default-first rule, the alias list of the secondary grader, the checker-wording test, and the two-way checker re-decision were specified after the first pool's analysis; each is reported as such. The second pool's model digests and licenses, including the non-commercial license of EXAONE 3.5, are listed in its manifest.

S8 Evaluation Methods and Prespecified Analyses

OVERTURN poses questions that a small fleet answers mostly right but often wrongly in unison, gives the agents a fixed corpus to cite, and grades every method against dataset gold. Its design meets the four requirements for evaluating arbitration set out in Section S6.1. Its protocol, items, grader, canonicalizer, analysis code, reducer binary, checker identity, and policy templates were frozen by hash before any confirmatory response was collected (Section S7).

Items. Gold is copied mechanically from four public datasets: SciFact claims with their abstracts (Wadden et al., 2020), PopQA functional relations in three popularity strata (Mallen et al., 2023), FEVER claims (Thorne et al., 2018), and TruthfulQA multiple-choice (Lin et al., 2022). The confirmatory set has 1,100 items: 250 SciFact, 400 PopQA, and 350 FEVER, which form the primary set P, and a 100-item TruthfulQA stratum T. A disjoint 160-item development split fixed every configuration and the sample size. TruthfulQA probes misconceptions that an

encyclopedic corpus rarely addresses. On development, the retrieved passages supported its gold on 18% of decisive items, against 56–90% elsewhere, so the protocol tests it separately, for non-inferiority.

Agent collection. Five small open-weight models (Llama 3.2 3B, Mistral 7B, Gemma 2 2B, Phi-3 mini, Qwen 2.5 3B), served locally with recorded digests at temperature 0.7 and a fixed seed, answer each question in two turns. In the first, each writes a search query of at most 12 words. Retrieval returns its top five passages. For PopQA, FEVER, and TruthfulQA it uses BM25 (Robertson and Zaragoza, 2009) over the FEVER 2017 Wikipedia passages. For SciFact it searches the abstracts with BM25 or a dense embedder (nomic-embed-text), chosen per item by a seeded draw. In the second turn each agent answers in a constrained output schema over a closed answer space (an entity, SUPPORTED/REFUTED, or an option letter). It also gives a confidence level and cites at most three of the passages it was shown. Requests are dispatched concurrently in a seeded random order, and arrival order is recorded. All 1,100 items were collected with all five answers parsed.

A second pool of five open-weight models from other families (Falcon 3 3B, OLMo 2 7B, EXAONE 3.5 7.8B, Nemotron-Mini 4B, Granite 3.3 2B), qualified on the development split before its own freeze, answered the same 1,100 items under the same protocol (main-paper second-pool results).

Answer representation. Each parsed answer becomes one AFFIRM contribution whose object is the canonical answer. Its confidence is 0.85, 0.65, or 0.45 for a stated high, medium, or low level, every agent has one calibration factor of 0.7, and each cited passage becomes one CITATION item of weight 1.0. All answers to an item share one slot, so each distinct canonical answer is a value of main-paper rule definition.

The checker sees each answer as a one-sentence proposition rendered without gold. For SciFact and FEVER this is the dataset claim, whose supports and refutes labels are swapped when read for the answer REFUTED. For TruthfulQA it is the question with the chosen option, and for PopQA a relation template filled with subject and answer.

Checker. Support labels come from Claude Sonnet 5.5 with tools disabled, a different model family from the agents, run with a pinned client whose version is part of the checker identity. It was validated on fresh FEVER and SciFact slices disjoint from every benchmark item (Table S6) and accepted into the policy before collection.

Validation limits. The checker emitted supports on 18 of 350 validation pairs whose gold relation was not supports. This raw-label rate was the pre-specified forecast input. It excludes false refutes: the raw validation records contain 22 refutes labels on 350 pairs whose gold relation was not refutes. A refutes label verifies a REFUTED answer after the answer-relative swap, so 18/350 is not a pooled false-verification rate for both answer directions.

Alternative checkers. Our current claim representation requires three-way labels: an answer REFUTED is verified by a refutes label on the original claim. A checker that only separates supported from unsupported cannot verify that answer through this representation. A binary checker could instead assess an explicitly negated proposition, but that would change the rendering and require separate validation; we did not test it. Substituting a compact two-way grounding checker (Tang et al., 2024) for the hosted checker on OVERTURN keeps overturns of right majorities low (5 against 4 on P) but lowers verified default first from 87.3% to 77.8%.

Table S6: Support-checker validation on fresh slices disjoint from OVERTURN. Intervals are bootstrap for F1 and precision and Wilson for proportions; thresholds apply to point estimates.

Metric	Result	95% CI	Threshold
macro-F1 (FEVER, $n = 300$)	0.899	[0.864, 0.932]	≥ 0.70
refutes precision (FEVER)	0.886	[0.819, 0.941]	≥ 0.80
supports precision (FEVER)	0.898	—	≥ 0.85
raw false-support rate (FEVER, SciFact)	18/350 (0.051)	[0.033, 0.080]	—
raw false-refute rate (post hoc audit)	22/350 (0.063)	—	—
prompt-injection success	0/50	[0%, 7.1%]	$\leq 5\%$
identical label on regeneration	48/50 (96%)	[86.5%, 98.9%]	$\geq 95\%$

Prompt sensitivity. We re-checked 227 pairs, including every pair that licensed an overturn or supported a wrong answer, under three paraphrased checker prompts. The paraphrases changed 2.6–5.7% of labels, against 4.4% for re-running the original prompt, and no wording changed the number of right majorities that verified default first overturns.

Methods. Every method decides the same slots from the same sealed inputs. Verify-then-vote is evaluated with three selectors. *Verified vote* uses plurality: the most common verified answer, or the majority when none is verified. The pre-specified primary method, which we call *verify-then-vote* below, uses the scoring reducer of Section S2 as its selector. It is the gate of main-paper rule definition over the order-free reducer (zero recency weight, recency removed from the tie-break voters, residual ties by identifier), with the evidence policy, a not-enough-information factor of 0.5, and the validated checker accepted. We chose it on development among nine gated configurations, as the one with the largest margin of corrected over overturned majorities. Ties were broken by fewer overturns and then by fewer policy settings changed from the base.

The *ungated reducer* uses the evidence policy without the gate, at the not-enough-information factor selected on development among ungated configurations (0.1). Evidence-only weights share its settings.

Verified default first keeps the majority when it is verified and otherwise applies verified vote. We specified it after the OVERTURN analysis, from its harm cases, and it is the pre-specified primary rule of the second pool (main-paper second-pool results).

Controls and model references. Controls isolate each ingredient: the order-free reducer with unverified citations (no evidence policy), evidence-only weights, *cite vote* (plurality among answers that cite any passage, unverified), confidence-weighted vote, and the best single model on development (Mistral).

Three Sonnet 5.5 references measure what a strong model achieves under different inputs. The judge sees the agents' answers, confidences, and cited passages and must pick one of the proposed answers. The reader answers from the union of the agents' retrieved passages, and the third reference is a closed-book model. A local judge, Mistral 7B with the same judge prompt, is the same-size alternative.

Tests. The protocol was frozen and hash-committed in the project repository before data collection; it was not posted to a public registry. The unit of analysis is a *decisive conflict*: an item whose agents give at least two distinct answers, at least one of them correct and one incorrect. An item is *contested* when its agents give at least two distinct canonical answers (622 items on P, 702 in all). The rest are decided without a model call, so cost and counterfactuals are reported over contested items. Majority vote is plurality over canonical answers with seeded random ties.

Table S7: Pre-specified tests. P: decisive conflicts on SciFact, PopQA, and FEVER; T: on the TruthfulQA stratum.

	Contrast	Test
H1a (primary)	verify-then-vote vs majority vote on P	exact two-sided McNemar, Holm with H1b, family-wise $\alpha = 0.05$
H1b	ungated reducer vs majority vote on P	as H1a
H3a	rate at which verify-then-vote overturns right majorities on P	one-sided 97.5% Clopper–Pearson (Clopper and Pearson, 1934) upper bound below 0.05
H3b	verify-then-vote vs majority vote on T	non-inferiority, Newcombe lower bound above -0.03
H4	harm bound of main-paper harm corollary	$c_D \leq F$ checked per item; F compared with $\hat{F}$ predicted from validation

Table S7 lists the pre-specified tests. Sample size followed a fixed rule from development discordance: 1,000 primary items, assuming the development decisive-conflict yield of 0.533 (about 533 eligible conflicts), give 0.93 power at $\alpha = 0.05$ (0.88 at Holm's (Holm, 1979) first-step $\alpha = 0.025$) for an accuracy difference of 0.075 at a planning discordance of 0.25.

S8.1 Second-pool protocol

A second pre-specified protocol tests verified default first on newly collected responses to the same questions and repeats every OVERTURN hypothesis unchanged. This tests transfer across agent pools, not generalization to unseen questions; the rule was designed after the first pool's analysis. A new pool of five open-weight agents, from developers and model families not in the first (Section S8), answers the same 1,100 items. A qualification rule on the development split fixed the pool. Each agent had to parse and finish reliably and cite on at least a quarter of its answers, and the pool had to produce enough conflicts. A pre-declared alternate replaced a candidate whose model server failed on every request. Protocol, analysis code, checker, judges, policy templates, and every hash were frozen before collection. The primary family is a fixed sequence for verified default first on P: superiority to majority vote (exact McNemar, $\alpha = 0.05$), then, only if that rejects, an overturn rate of right majorities with one-sided 97.5% upper bound below 0.05. The replication family repeats H1a, H1b, H3a, and H3b with the same tests. A non-binding futility look at 300 items read only the primary corrected and overturned counts (22 against 1).

S8.2 Historical forecast and calibration limits

The gate's exact counting condition, $c_D \leq F$, does not require a statistical model. Predicting checker-caused false verification does. A calibrated model would distinguish answer correctness from passage entailment. Let k_i^+ and k_i^- count checked pairs whose passages do not actually license the candidate answer, in the raw-support and raw-refute directions respectively. Under independent errors with transportable direction-specific rates r_+ and r_-, the expected number of questions with at least one false verification is

$$\sum_i \left[1 - (1 - r_+)^{k_i^+} (1 - r_-)^{k_i^-}\right].$$

The independently counted validation rates are $r_+ = 18/350$ and $r_- = 22/350$, on different gold-relation subsets of the same validation records. They do not by themselves establish transportability to the deployment passages, renderings, or answer types. Replacing one rate with the other is therefore insufficient to recalibrate deployment harm.

The released pre-specified calculation instead uses $\hat{F} = \sum_i [1 - (1 - r)^{k_i}]$, with $r = 18/350$ and k_i the number of checked pairs on gold-incorrect values. It gives 52.7 on P and 9.9 on T,

compared with 56 and 4 items with a verified gold-incorrect answer. Its reported predictive intervals on P are [40, 66] with fixed r and [30, 82] with uncertain r. These intervals reflect the assumed model; they do not resolve the mismatch between gold-incorrect answers and non-entailing passages, or the omitted false-refute channel. The exploratory form substitutes checks on non-default values and gives 60.3, with interval [47, 75]. That exposure also includes genuinely supporting passages and is not an adjudicated false-verification population.

We retain these calculations to document the original analysis, not as calibrated forecasts or empirical upper bounds on harm. Entity mismatch, false sources, and gold disagreement can produce verified gold-incorrect answers without raw checker misclassification. The 24 model-attributed checker-error items are not an independently validated calibration target. A valid comparison needs passage-level assessment across the relevant exposure population, not only the selected 82 verified wrong pairs, together with direction-specific error rates and justified assumptions about dependence. The artifact's validation audit and blank human-review packet make these outstanding requirements explicit.

S8.3 Pooled harm rate and dependence sensitivity

This secondary analysis concerns the original scored rule, not verified default first. It was specified after the first pool's harm result was known and before the second pool was collected. For pool j, let q_j be the probability that the scored rule returns an incorrect answer conditional on a decisive primary conflict whose reported majority baseline is correct. These are losses relative to the seeded-tie majority baseline, not the rule's own deterministic default. The observed counts are $(c_1, n_1) = (12, 376)$ and $(c_2, n_2) = (8, 400)$. The target is the observed-denominator-weighted mean of the two pool-specific rates,

$$\bar{q} = \frac{376q_1 + 400q_2}{776}, \qquad \widehat{\bar{q}} = \frac{12 + 8}{376 + 400} = 2.5773\%.$$

It describes this mixture of the two evaluated pools; it is not an estimate over a population of arbitrary future agent pools.

Treating the 776 outcomes as one independent, common-rate binomial sample gives the one-sided 97.5% Clopper–Pearson (Clopper and Pearson, 1934) upper bound 3.9525%. Shared questions and reused checker labels can violate that independence model. The reported 5.01% sensitivity instead combines two marginal bounds using Bonferroni. Set the total tail probability to $\alpha = 0.025$, allocate $\alpha/2 = 0.0125$ to each pool, and compute one-sided 98.75% Clopper–Pearson upper limits

$$U_j = \mathrm{Beta}^{-1}(0.9875;\, c_j + 1, n_j - c_j), \qquad U_1 = 0.0587031, \quad U_2 = 0.0420556.$$

Their weighted upper limit is

$$U = \frac{376U_1 + 400U_2}{776} = 0.0501219 \approx 5.01\%.$$

If both marginal limits cover, then $\bar{q} \le U$. By the union bound, the probability that either fails is at most $0.0125 + 0.0125 = 0.025$, without assuming independence between pools. The construction therefore permits different pool-specific rates and arbitrary cross-pool dependence, including dependence from shared items or labels. It does not estimate that dependence, and 5.01% is not a cluster-bootstrap percentile.

The coverage statement still requires valid marginal binomial limits: within each pool, the eligible item outcomes are modeled as independent Bernoulli trials with a common conditional harm probability. It does not protect against arbitrary within-pool item dependence, changing the selection rule after observing the analyzed outcomes, or errors in the gold labels. The weights are the observed correct-majority denominators, rather than a claim about how often

future pools will occur. Because this pooling plan followed the first-pool result, we report it as a secondary sensitivity analysis, not an additional independent confirmation. The resulting upper limit is slightly above the 5% criterion, so the pooled low-harm claim is not established under this sensitivity.

The artifact's `real-agent-eval/v6/pooled_h3a_v6.py` computes these limits from the two per-item decision tables and first verifies their counts against the evaluation summaries. Its separate item-cluster bootstrap retains both pools' outcomes together when resampling each of the 1,000 primary questions (10,000 resamples, seed 42); that descriptive two-sided 95% interval, [1.50%, 3.85%], is a different calculation and does not supply the 5.01% bound.

S9 Sealed Evidence and Replay

Verify-then-vote is useful to an auditor only if its inputs are fixed: which passage each agent cited, what the passage said, and what the checker labeled it. Everything the decision reads is therefore produced first and sealed into one content-addressed artifact set, and the decision is a pure function of that set and a hashed policy.

S9.1 Capture and support checking

Capture turns each cited source into a record. Two kinds of source are supported. Web citations (URLs, DOIs, RFC numbers, arXiv identifiers) are fetched once per epoch with TLS verification, bounded redirects, per-host rate limits, and robots.txt checks. Capture refuses private and metadata addresses, and each fetch gets one of 14 outcome classes. Corpus citations name a passage of a fixed, hashed retrieval corpus by a canonical identifier, and capture reads the passage from it. Each record holds the outcome, the body's content hash, and at most three 1,000-character spans, and is identified by a hash of its canonical form; bodies stay in a cache outside A.

Support checking asks an LLM to label each (claim, spans) pair as supports, refutes, or not-enough-information. The claim is rendered as a one-sentence proposition, and the checker sees only that proposition and the passage, never other agents' answers, vote counts, or gold. Checks are de-duplicated by (proposition, passage), so each label is computed once and reused by every policy that reads it. The checker's identity is a hash of its model, prompt, tool version, and flags, and fetched text is treated as data, with delimiter look-alikes neutralized.

Both record types are sealed into A's evidence store. A label applies only if its checker identity is on the policy's list of accepted checkers. That list is part of the hashed policy, so the reducer enforces acceptance-before-use; validating a checker before accepting it is the study or application's responsibility. Section S2 gives the evidence weights the scoring selector derives from captures and labels.

Replay reads only the sealed records, so evidence verification inherits the replay guarantee below. A flipped label changes the root hash, and removing an accepted checker changes the policy hash.

S9.2 Artifact set, policy, and certificate

The artifact set A contains the normalized intent, the contributions, the evidence items, the calibration snapshot (one factor per agent), and optionally an evidence store of capture and support records (Section S9.1). Each contribution carries its content-addressed identifier and recorded arrival index σ. Each evidence item records its type, source, weight, verification level, and agent-reported reachability. A is serialized canonically, without wall-clock timestamps, and hashed before reduction. Every field the reducer reads is inside that hash.

The policy P holds the weights, lowering factors, deadband, Schulze indifference threshold, arity and exclusivity tables, and optional tie-break, candidate, and evidence settings, and is

frozen per compilation. Its hash also covers the constants compiled into the reducer, so, under the same reducer implementation, equal policy hashes identify the same scoring settings, subject to hash-collision assumptions. The policy hash does not by itself bind implementation identity; reproductions must also identify the reducer source or binary. The *certificate* hashes A's root hash, the policy hash, and the admitted identifiers in admission order.

The purity constraints are few. Calibration comes only from the snapshot in A. All ordering derives from canonical keys or the recorded σ alone, independent of iteration order and the clock. Scores are compared only after quantization to integers on a 10,000 scale. These comparisons use integers, but agreement across platforms still depends on the floating-point arithmetic before quantization; the cross-build tests in main-paper replay tests establish agreement for the tested inputs and builds. Under them the reducer satisfies one guarantee.

Guarantee 1 (Replay)**.** *The reducer is a pure function:* $A_1 = A_2 \wedge P_1 = P_2 \implies C(A_1, P_1) = C(A_2, P_2)$*. Its output is the admitted/rejected partition of contribution identifiers together with the per-slot selections (slot, members, winner, and quantized scores). Reduction begins only after* A *is sealed.*

S9.3 Why seal

Replay needs only exact preservation of the reducer's inputs: exact input logging with *any* deterministic reducer, weighted voting included, would replay. The reducer hashes its inputs for three further properties: tamper evidence for an artifact that travels between parties, cheap equality checks, and idempotent re-submission. The architectural principle is that arbitration is a function over an explicitly preserved, canonical input. Anything that can vary belongs to artifact production and is recorded before it can influence reduction.

Two consequences follow. Arrival order is a recorded input: recency reads σ from A, so even an order-dependent feature depends only on sealed state. Whether arrival order *should* matter is a policy question, and Section S6.1 makes the case for order-free selection. Model calls are artifacts: their outputs are committed into A, and the reducer is deterministic over those recorded outputs. Re-running a model to *regenerate* an artifact is a separate operation that lies outside the replay guarantee.

Tamper evidence is relative to a trusted commitment. Subject to hash-collision assumptions, hashes make an edit detectable against a certificate or root hash that the verifier already trusts. A producer who hands an auditor a decision also records its certificate somewhere the auditor trusts independently of the bundle, such as a signed release, an append-only log, or a preregistration's freeze record. Any bundle presented later is checked against that record, and an edited claim, label, arrival index, policy, or output then fails (main-paper replay tests). An adversary who can replace the bundle and every trusted copy of its certificate can recompute all hashes, so the guarantee is as strong as the authentication of the certificate. For OVERTURN, the preregistration's freeze record fixes the analysis code, reducer binary, checker identity, and policy templates by hash before collection. The artifact includes the root hashes and certificates of the collected sets. A verifier must preserve a trusted copy independently; inclusion in the same editable bundle does not authenticate that bundle. The certificate keeps 64 bits; an auditor who needs collision resistance against the producer records the 128-bit root and policy hashes instead.

S10 Replay and Tamper Tests

We tested Guarantee 1 at three levels on five synthetic topics (100 agents each, seed 42). In memory, all 500 recompilations (100 per topic) gave identical certificates, and 20 random permutations of the input order per topic (100 in all) left root hash, certificate, and winners unchanged.

Table S8: Cross-build replay matrix: topics passing out of five, for each replaying build (rows) and producing build (columns). Dumps were byte-identical across builds.

	producer			
replayer	macOS arm64	macOS x86-64 (translated)	Linux arm64	Linux x86-64 (translated)
macOS arm64	5/5	5/5	5/5	5/5
macOS x86-64 (translated)	5/5	5/5	5/5	5/5
Linux arm64	5/5	5/5	5/5	5/5
Linux x86-64 (translated)	5/5	5/5	5/5	5/5

Table S9: Tamper cases (each replayed on four builds; all 36 rejected identically).

Mutation	Outcome
Edit a claim, keep identifier and root hash	root-hash mismatch
Edit a claim, recompute root hash, keep identifier	identifier differs from its content hash
Insert a SYNTHESIS contribution with valid identifier and root hash	rejected at validation
Edit a claim, recompute identifier and root hash, keep stored outputs	outputs differ
Change one contribution's arrival index	root-hash mismatch
Swap a stored winner	outputs differ
Change the deadband in the policy	policy mismatch, rejected before reduction
Remove the identifier scheme from the sealed set	root-hash mismatch
Relabel the artifact format	rejected as unsupported

From file, each sealed set, policy, and output was written out, reloaded, verified against its hashes, reduced again, and compared. Across builds, four builds of the reducer (macOS arm64 native, macOS x86-64 translated, Linux arm64 in a container, and Linux x86-64 translated in a container) each produced and replayed every topic. They span two instruction sets, two operating systems, and two compiler patch releases, all on one physical host. The 20 dumps were byte-identical per topic across builds, and all 80 cross-replays passed (Table S8).

Tamper evidence is the other half of the boundary's value, measured against the stored certificate and outputs as the trusted reference (Section S9.3). Nine forgeries of one sealed topic, each replayed on all four builds, were all rejected identically (36 of 36; Table S9). The informative cases are the consistent forgeries. An edited claim with a recomputed root hash fails because its identifier differs from its content hash. An inserted SYNTHESIS contribution with valid hashes fails validation, and an edit that recomputes every hash but keeps the stored outputs fails on output comparison. A forgery that also replaces the stored outputs and certificate is caught only against an independently held copy of the original certificate, which is why the certificate is recorded outside the bundle. Flipping a ground-truth label passes, as it should, because ground-truth labels lie outside the reducer's input.

Further tests cover 12 other reducer-read fields (root hash changes), a changed weight or exclusivity entry (policy mismatch), and a changed policy with a rewritten hash (certificate mismatch). The main-paper limitations section states the scope of these results.

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